Exponential and Logarithmic Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exponential equations**

Solve $3^{x+1} = 81$ exactly.

$3^{x+1} = 3^4$, so $x + 1 = 4$ and $x = 3$.

2 **[Foundational]** • **Target: Exponential equations**

Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.

$2x = \log_5 17 = \frac{\ln 17}{\ln 5}$, so $x = \frac{\ln 17}{2\ln 5} \approx 0.880$.

3 **[Foundational]** • **Target: Logarithmic equations**

Solve $\log_2 x + \log_2 (x - 6) = 4$, checking for extraneous solutions.

$\log_2 (x(x-6)) = 4$, so $x^2 - 6x - 16 = 0$, giving $(x - 8)(x + 2) = 0$. Reject $x = -2$ (logarithm of a negative number): $x = 8$.

4 **[Foundational]** • **Target: Logarithmic equations**

The pH of a solution is $\text{pH} = -\log[\text{H}^+]$. Find the hydrogen-ion concentration $[\text{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.

$[\text{H}^+] = 10^{-4.6} \approx 2.5 \times 10^{-5}$ mol/L.

5 **[Foundational]** • **Target: More exponential equations**

Solve $2^x = 3^{x-1}$, giving an exact expression and a decimal to three places.

Take logs: $x\ln 2 = (x - 1)\ln 3$, so $x(\ln 2 - \ln 3) = -\ln 3$, giving $x = \frac{\ln 3}{\ln 3 - \ln 2} = \frac{\ln 3}{\ln(3/2)} \approx \frac{1.0986}{0.4055} \approx 2.710$.

6 **[Foundational]** • **Target: More exponential equations**

Solve $4^x - 3 \cdot 2^x - 4 = 0$ exactly, using the substitution $u = 2^x$.

$4^x = (2^x)^2 = u^2$: $u^2 - 3u - 4 = 0$, so $(u - 4)(u + 1) = 0$. $u = 4$ (rejecting $u = -1$, since $2^x > 0$): $2^x = 4$, so $x = 2$.

7 **[Foundational]** • **Target: Systems and applications**

A bacteria culture grows according to $N(t) = 500e^{kt}$. If the population reaches 2000 after 6 hours, find k (4 d.p.) and hence the time for the population to reach 10,000, to one decimal place.

$2000 = 500e^{6k}$: $e^{6k} = 4$, so $k = \frac{\ln 4}{6} \approx 0.2310$. For $N = 10000$: $10000 = 500e^{0.2310t}$, so $e^{0.2310t} = 20$, $t = \frac{\ln 20}{0.2310} \approx 12.97 \approx 13.0$ hours.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Systems and applications**

Solve the equation $e^x + e^{-x} = 3$ exactly, using the substitution $u = e^x$ (this expression is related to $2\cosh x$).

Multiply by $u = e^x$: $u^2 - 3u + 1 = 0$, so $u = \frac{3 \pm \sqrt{5}}{2}$. Both roots are positive, giving $x = \ln\left(\frac{3+\sqrt{5}}{2}\right)$ or $x = \ln\left(\frac{3-\sqrt{5}}{2}\right)$.

9 **[Intermediate]** • **Target: Combined logarithmic and exponential practice**

Solve $9^x = 27^{x-1}$ exactly, writing both sides with base 3.

$3^{2x} = 3^{3(x-1)}$: $2x = 3x - 3$, so $x = 3$.

10 **[Intermediate]** • **Target: Equations requiring both exponential and logarithmic techniques**

Solve $e^{\ln x + \ln 2} = 10$ for x, simplifying the exponent first.

$e^{\ln x + \ln 2} = e^{\ln(2x)} = 2x$ (since $e^{\ln u} = u$). So $2x = 10 \Rightarrow x = 5$.

11 **[Intermediate]** • **Target: Verifying solutions numerically**

For the equation $5^{2x} = 17$ solved earlier ($x \approx 0.880$), verify the solution by substituting back and computing both sides to 2 decimal places.

$5^{2(0.880)} = 5^{1.76} \approx 17.02$, matching the right side 17 (small rounding difference due to the decimal approximation of x), confirming the solution.

12 **[Intermediate]** • **Target: Equations with different bases on each side**

Solve $10^{2x-1} = 500$, giving an exact logarithmic expression and a decimal to three places.

$2x - 1 = \log 500$, so $x = \frac{1 + \log 500}{2} \approx \frac{1 + 2.6990}{2} \approx 1.850$.

13 **[Intermediate]** • **Target: A modeling problem requiring both equation types**

A population model is $P(t) = 800e^{0.05t}$. Find the time t when $P(t) = 2000$, to two decimal places, and separately find $P^{-1}(1500)$ (the time when the population reaches 1500), to two decimal places.

For $P = 2000$: $2000 = 800e^{0.05t} \Rightarrow 2.5 = e^{0.05t} \Rightarrow t = \frac{\ln 2.5}{0.05} \approx 18.33$. For $P = 1500$: $1500 = 800e^{0.05t} \Rightarrow 1.875 = e^{0.05t} \Rightarrow t = \frac{\ln 1.875}{0.05} \approx 12.58$.

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14 **[Intermediate]** • **Target: Checking solutions by substitution**

Verify the solution $x = 8$ found earlier for $\log_2 x + \log_2 (x - 6) = 4$ by substituting back into the original equation.

$\log_2 (8) + \log_2 (8 - 6) = \log_2 (8) + \log_2 (2) = 3 + 1 = 4$ ✓, confirming $x = 8$ satisfies the original equation.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Checking solutions by substitution** • **Cross-Lesson: 3.01 The Unit Circle and Exact Values**

A radioactive sample decays according to $A(t) = A_0 e^{-0.08t}$. If the initial amount is 500 g, find the time t (to two decimal places) at which only 50 g remains, and separately find the time at which 90% of the original sample has decayed away.

For $A = 50$: $50 = 500e^{-0.08t} \Rightarrow 0.1 = e^{-0.08t} \Rightarrow t = \frac{\ln 0.1}{-0.08} \approx 28.79$. For 90% decayed, 10% remains ($A = 50$ g as well, since 10% of 500 is 50): this is the same equation, so $t \approx 28.79$ — confirming both descriptions refer to the same instant.

16 **[Challenging]** • **Target: Equations requiring both exponential and logarithmic techniques** • **Cross-Lesson: 3.02 Fundamental Trigonometric Identities**

Level 2 · MediumSolve the logarithmic equation $\log_2 (x + 3) + \log_2 (x - 1) = 5$ analytically for all real solutions. Show the verification step that eliminates extraneous roots.

Step 1 (Apply Product Property): $\log_2 ((x + 3)(x - 1)) = 5.$ Step 2 (Convert to Exponential Form): $(x + 3)(x - 1) = 2^5 = 32.$ Step 3 (Solve the Quadratic Equation): $x^2 + 2x - 3 = 32 \Rightarrow x^2 + 2x - 35 = 0.$ $(x + 7)(x - 5) = 0 \Rightarrow x = -7$ or $x = 5.$ Step 4 (Domain Verification): For $x = -7$: $x - 1 = -8 \leq 0$, and $\log_2 (-8)$ is undefined in the real numbers. Hence $x = -7$ is an extraneous solution. For $x = 5$: $5 + 3 = 8 > 0$ and $5 - 1 = 4 > 0$. Checking.

17 **[Challenging]** • **Target: Verifying solutions numerically** • **Cross-Lesson: 3.03 Sum and Difference Formulas**

Find all real solutions to the rational equation $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$.

Notice that the common denominator factors as $x^2 + x - 6 = (x - 2)(x + 3)$. Domain restrictions: $xeq2$ and $xeq -3$. Multiply both sides by $(x - 2)(x + 3)$: $x(x + 3) + 1(x - 2) = 5 \Rightarrow x^2 + 3x + x - 2 = 5$ $x^2 + 4x - 7 = 0$ Apply the quadratic formula with $a = 1, b = 4, c = -7$:

$x = \frac{-4 \pm \sqrt{4^2 - 4(1)(-7)}}{2(1)} = \frac{-4 \pm \sqrt{16 + 28}}{2} = \frac{-4 \pm \sqrt{44}}{2} = \frac{-4 \pm 2\sqrt{11}}{2} = \mathbf{-2 \pm \sqrt{11}}$ Neither solution equals 2 or -3, so both solutio.

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18 [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Solve the exponential equation $4^{x+1} - 5(2^x) + 1 = 0$ for all real values of x.

Rewrite the first term in terms of base 2: $4^{x+1} = 4^1 \cdot (2^2)^x = 4 \cdot (2^x)^2$. Substitute $u = 2^x$ (with constraint $u > 0$): $4u^2 - 5u + 1 = 0$. Factor the quadratic: $(4u - 1)(u - 1) = 0 \Rightarrow u = \frac{1}{4}$ or $u = 1$. Back-substitute $2^x = u$:$2^x = \frac{1}{4} = 2^{-2} \Rightarrow x = -22^x = 1 = 2^0 \Rightarrow x = 0$Both solutions are real and valid: $x = -2$ and $x = 0$.

19 [Challenging] • Target: A modeling problem requiring both equation types • Cross-Lesson: 4.01 Circles and Parabolas

Solve the logarithmic equation $\log_4(x + 6) + \log_4(x - 6) = 3$, checking for extraneous roots.

Step 1: Domain restriction: We require $x + 6 > 0$ and $x - 6 > 0$, so $x > 6$. Step 2: Condense logarithms using the product rule: $\log_4((x + 6)(x - 6)) = 3 \Rightarrow x^2 - 36 = 4^3 = 64$. Step 3: Solve for x and filter by domain: $x^2 = 100 \Rightarrow x = \pm 10$. Reject $x = -10$ since $-10ot > 6$. The unique valid solution is **$x = 10$**.

20 [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 4.02 Ellipses

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Exponential equations

What is the sum of all real solutions to the equation $2^{\log_5(x^2)} - 6 \cdot x^{\log_5(2)} + 8 = 0$?

a. 30 ✓

b. 26

c. 6

d. 3

To solve the equation for real x, we first determine the domain and apply fundamental logarithmic identities. 1. **Domain:** The term $x^{\log_5(2)}$ requires $x > 0$ for the power to be well-defined over the real numbers. For $x > 0$, $\log_5(x^2)$ is also well-defined. 2. **Logarithmic and Exponential Identities:** Using the power rule for logarithms with $x > 0$: $\log_5(x^2) = 2\log_5(x)$ Thus, the first term can be rewritten as: $2^{\log_5(x^2)} = 2^{2\log_5(x)} = (2^{\log_5(x)})^2$ Next, recall t.

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22 [Foundational] • Target: Logarithmic equations

The rate of change of a quantity N is proportional to its current value, modeled by the differential equation $\frac{dN}{dt} = kN$, where k is a constant. If the initial population at time $t = 0$ is N_0, and at time $t = 5$ years, the population has grown to $2N_0$, what is the population at time $t = 15$ years, expressed in terms of N_0?

a. $6N_0$

b. $8N_0$ ✓

c. $7N_0$

d. $10N_0$

The differential equation $\frac{dN}{dt} = kN$ is a standard model for exponential growth. The solution to this differential equation is of the form $N(t) = N_0e^{kt}$, where N_0 is the initial population at $t = 0$. We are given that at $t = 5$ years, the population is $2N_0$. We can use this information to find the constant k. $N(5) = N_0e^{k \cdot 5} = 2N_0$ Divide both sides by N_0: $e^{5k} = 2$ Take the natural logarithm of both sides: $5k = \ln(2)$ So, $k = \frac{\ln(2)}{5}$. Now we need to find the popul.

23 [Foundational] • Target: More exponential equations

The equation $\log_a x = b$ and $\log_b x = a$ simultaneously implies:

a. No simultaneous solution

b. $a = b$

c. $x = 0$

d. $x = a^b = b^a$ ✓

Applying Exponential & Logarithmic Functions principles, Both define x: $x = a^b$ and $x = b^a$. Therefore, the verified correct option is $x = a^b = b^a$.

24 [Foundational] • Target: Systems and applications

Find the number of solutions to $e^x = x + 2$.

a. 1

b. 2 ✓

c. Infinitely many

d. 0

Graph: e^x convex, $x + 2$ linear — they intersect at 2 points for this particular line.

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25 [Foundational] • Target: Combined logarithmic and exponential practice

Solve $e^x = x + 2$ (number of solutions).

a. 0 solutions

b. **2 solutions (one for $x > 0$, one for $x < 0$) ✓**

c. 1 solution

d. Infinitely many

e^x and $x + 2$ intersect at 2 points (verify graphically or analytically).

26 [Foundational] • Target: Equations requiring both exponential and logarithmic techniques

Find all solutions to $4^x - 6 \cdot 2^x + 8 = 0$.

a. $x = 0, 1$

b. $x = 2, 4$

c. $x = 3, 4$

d. $x = 1, 2$ ✓

$(2^x)^2 - 6(2^x) + 8 = 0$. Let $u = 2^x$: $(u - 2)(u - 4) = 0 \Rightarrow x = 1, 2$.

27 [Foundational] • Target: Verifying solutions numerically

Find the number of real solutions to $\ln x = x^2 - 3$.

a. 3

b. 1

c. 0

d. 2 ✓

Graph intersection: $\ln x$ and $x^2 - 3$ cross at 2 points.

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28 [Foundational] • Target: Equations with different bases on each side

The graph of the function $f(x) = a \cdot 2^{x-h} + k$ is shown below. The graph passes through the points $(2, 9)$ and $(4, 33)$. The horizontal asymptote is $y = 1$. Find the value of a.

a. **4** ✓

b. 2

c. 1/2

d. 8

The function is given by $f(x) = a \cdot 2^{x-h} + k$. The horizontal asymptote is $y = 1$, which means $k = 1$. So, the function can be written as $f(x) = a \cdot 2^{x-h} + 1$. We are given two points on the graph: $(2, 9)$ and $(4, 33)$. Using the point $(2, 9)$: $9 = a \cdot 2^{2-h} + 1$ $8 = a \cdot 2^{2-h}$ (*) Using the point $(4, 33)$: $33 = a \cdot 2^{4-h} + 1$ $32 = a \cdot 2^{4-h}$ (**) Now we have a system of two equations with two unknowns, a and h: 1) $8 = a \cdot 2^{2-h}$ 2).

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A modeling problem requiring both equation types

Consider the function $f(x) = 2^x$. Which of the following points lies on the graph of $f(x)$?

a. $(1, 1)$

b. $(2, 4)$ ✓

c. $(3, 6)$

d. $(0, 2)$

The function is given by $f(x) = 2^x$. We need to find which of the given points satisfies this equation. For the point $(1, 1)$: $f(1) = 2^1 = 2$. Since $2eq1$, this point is not on the graph. For the point $(2, 4)$: $f(2) = 2^2 = 4$. Since $4 = 4$, this point is on the graph. For the point $(3, 6)$: $f(3) = 2^3 = 8$. Since $8eq6$, this point is not on the graph. For the point $(0, 2)$: $f(0) = 2^0 = 1$. Since $1eq2$, this point is not on the graph. Therefore, the point $(2, 4)$ lies on the graph of.

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30 [Intermediate] • Target: Checking solutions by substitution

The graph of the exponential function $f(x) = 2^x$ is given. A point P on the graph has a y-coordinate of 8. What is the x-coordinate of point P?

a. 1

b. 2

c. **3** ✓

d. 4

The function is given by $f(x) = 2^x$. We are looking for a point P on the graph where the y-coordinate is 8. This means we need to solve for x when $f(x) = 8$. So, we have the equation $2^x = 8$. Since $8 = 2^3$, we can rewrite the equation as $2^x = 2^3$. By equating the exponents, we find that $x = 3$. Therefore, the x-coordinate of point P is 3.

31 [Intermediate] • Target: Exponential equations

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k,$ where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|.$ Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$ ✓

b. $k = -2;\ x = 7$

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

1. **Find the parameter k:** By the definition of an inverse function, $f^{-1}(4) = 2 \Leftrightarrow f(2) = 4$. Substituting $x = 2$ into $f(x)$: $f(2) = 3(2) + k = 6 + k\ 6 + k = 4 \Rightarrow k = -2$ 2. **Determine the inverse function $f^{-1}(x)$:** With $k = -2$, $f(x) = 3x - 2$. Setting $y = 3x - 2$ and solving for x: $3x = y + 2 \Rightarrow x = \frac{y+2}{3}$ Thus, $f^{-1}(x) = \frac{x+2}{3}$. 3. **Solve the equation $f^{-1}(x) = g(x)$:** $\frac{x+2}{3} = |x - 4| \Leftrightarrow x + 2 = 3|x - 4|$.

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32 **[Intermediate]** • **Target: Logarithmic equations**

The bar graph shows the distribution of the number of goals scored per game by a soccer team across all 40 games of its regular season. If the team plays k post-season games and scores exactly 3 goals in each of those games, the mean number of goals scored per game across all regular-season and post-season games combined will be exactly 2. What is the value of k?

a. 8

b. 12

c. 20

d. **24** ✓

To find k, first determine the total number of goals scored during the regular season from the bar graph: - Games with 0 goals: $8 \Rightarrow 0 \times 8 = 0$ goals - Games with 1 goal: $14 \Rightarrow 1 \times 14 = 14$ goals - Games with 2 goals: $12 \Rightarrow 2 \times 12 = 24$ goals - Games with 3 goals: $6 \Rightarrow 3 \times 6 = 18$ goals Total regular-season games: $8 + 14 + 12 + 6 = 40$. Total regular-season goals: $0 + 14 + 24 + 18 = 56$. When the team plays k additional games scoring 3 goals in each.

33 **[Intermediate]** • **Target: More exponential equations**

Consider the equation $\sqrt[3]{25 - x^2} + \sqrt[3]{x^2 + 3} = 4$ What is the product of all real solutions to this equation?

a. -24 ✓

b. 24

c. -48

d. 0

To solve the equation, let $u = 25 - x^2$ and $v = x^2 + 3$. Observe that the sum of the radicands eliminates x^2: $u + v = (25 - x^2) + (x^2 + 3) = 28$ The original equation can be written in terms of rational exponents: $u^{1/3} + v^{1/3} = 4$ Let $a = u^{1/3}$ and $b = v^{1/3}$. Then: $a + b = 4 \quad \text{and} \quad a^3 + b^3 = 28$ Using the algebraic identity $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$: $28 = 4^3 - 3ab(4)$ $28 = 64 - 12ab$ $12ab = 36 \Rightarrow ab = 3$ Since $a + b = 4$ and $ab = 3$, a and.

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34 [Intermediate] • Target: Systems and applications

Consider the function $f(x) = \log_3(ax + b)$, where a and b are positive constants. If the graph of $y = f(x)$ passes through the points $(1, 2)$ and $(10, 3)$, find the value of a.

a. 3

b. **1** ✓

c. 1/3

d. 9

The function is given by $f(x) = \log_3(ax + b)$. We are given two points that lie on the graph of this function: $(1, 2)$ and $(10, 3)$. Substituting the first point $(1, 2)$ into the function, we get: $2 = \log_3(a(1) + b)$ $2 = \log_3(a + b)$ By the definition of logarithms, this can be rewritten in exponential form as: $3^2 = a + b$ $9 = a + b$ (Equation 1) Substituting the second point $(10, 3)$ into the function, we get: $3 = \log_3(a(10) + b)$ $3 = \log_3(10a + b)$ In exponential form, this is: $3^3 = 10a + b$.

35 [Intermediate] • Target: Combined logarithmic and exponential practice

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

a. $y = -5$ ✓

b. $y = 0$

c. $y = 3$

d. $x = -1$

An exponential function in the standard form $f(x) = a \cdot b^{x-h} + k$ (with base $b > 1$) has a horizontal asymptote determined by its vertical shift, $y = k$. As $x \to -\infty$, the exponential power $2^{x+1} \to 0$. Substituting this into the function yields: $\lim_{x \to -\infty} f(x) = 3(0) - 5 = -5$ Thus, the horizontal asymptote is the horizontal line $y = -5$. Common Misconceptions: - $y = 0$: Assumes the asymptote is always the x-axis, forgetting the vertical shift by -5..

36 [Intermediate] • Target: Equations requiring both exponential and logarithmic techniques

The graph of the function $g(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(3, 24)$. If a and b are positive constants, what is the value of a?

a. 1.5

b. 2

c. **3** ✓

d. 6

The function is given by $g(x) = a \cdot b^x$. Since the graph passes through $(1, 6)$ and $(3, 24)$, we can set up a system of equations: 1. $6 = a \cdot b^1$ 2. $24 = a \cdot b^3$ From equation (1), we can express a in terms of b: $a = \frac{6}{b}$. Substitute this expression for a into equation (2): $24 = \left(\frac{6}{b}\right) \cdot b^3$ Simplify the equation: $24 = 6 \cdot b^{3-1}$ $24 = 6 \cdot b^2$ Now, solve for b^2: $b^2 = \frac{24}{6}$ $b^2 = 4$ Since b is a positive constant, we take the po.

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37 [Intermediate] • Target: Verifying solutions numerically

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

a. **a = 4, b = 3 ✓**

b. a = 3, b = 4

c. a = 6, b = 2

d. a = 2, b = 6

Let the function be $f(x) = a \cdot b^x$. We are given that the graph passes through $(1, 12)$ and $(3, 108)$. Substituting these points into the function equation, we get two equations: 1) $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2) $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this expression for a into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12 \cdot b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b must be positive for an exponen.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a.

a. 2

b. 3

c. **4 ✓**

d. 6

Since the graph of $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$, we can set up a system of two equations: 1. $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2. $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b is the base of an exponential function, $b > 0$. Thus, $b = 3$. Now, substitute the value of b back into equ.

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39 [Challenging] • Target: A modeling problem requiring both equation types • Cross-Lesson: 4.01 Circles and Parabolas

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(2, 36)$. If $b > 0$ and $beq1$, find the value of a.

a. **4** ✓

b. 3

c. 12

d. 1

Let the function be $f(x) = a \cdot b^x$. We are given that the graph passes through $(1, 12)$ and $(2, 36)$. Using the point $(1, 12)$, we have: $12 = a \cdot b^1$ (Equation 1) Using the point $(2, 36)$, we have: $36 = a \cdot b^2$ (Equation 2) Divide Equation 2 by Equation 1: $\frac{36}{12} = \frac{a \cdot b^2}{a \cdot b}$ $3 = b$ Now substitute the value of b into Equation 1: $12 = a \cdot 3$ $a = \frac{12}{3}$ $a = 4$ Thus, the value of a is 4.

40 [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 4.02 Ellipses

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(2, 18)$. What is the value of b?

a. 2

b. **3** ✓

c. 4

d. 6

The function is given by $f(x) = a \cdot b^x$. We are given two points on the graph: $(1, 6)$ and $(2, 18)$. Substituting the first point into the function, we get: $6 = a \cdot b^1$ (Equation 1) Substituting the second point into the function, we get: $18 = a \cdot b^2$ (Equation 2) To find b, we can divide Equation 2 by Equation 1: $\frac{18}{6} = \frac{a \cdot b^2}{a \cdot b}$ $3 = b$ Therefore, the value of b is 3.

41 [Challenging] • Target: Exponential equations • Cross-Lesson: 4.03 Hyperbolas

The graph of the exponential function $f(x) = a \cdot 2^x$ passes through the point $(3, 40)$. What is the value of a?

a. $\frac{5}{2}$ ✓

b. 5

c. 10

d. 20

The function is given by $f(x) = a \cdot 2^x$. We are given that the graph passes through the point $(3, 40)$. This means that when $x = 3$, $f(x) = 40$. We can substitute these values into the function equation: $40 = a \cdot 2^3$ Calculate 2^3: $2^3 = 2 \cdot 2 \cdot 2 = 8$ So the equation becomes: $40 = a \cdot 8$ To solve for a, divide both sides by 8: $a = \frac{40}{8}$ $a = 5$ Therefore, the value of a is 5.

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42 [Challenging] • Target: Logarithmic equations • Cross-Lesson: 5.01 Vectors: Components and Operations

Consider the matrix equation: $(3, -1; 2, k)(x; y) = (5; 7)$. If the matrix $(3, -1; 2, k)$ is singular, what is the value of k?

a. $-3/2$ ✓

b. $3/2$

c. $-2/3$

d. $2/3$

A matrix is singular if and only if its determinant is zero. The determinant of a 2x2 matrix $(a, b; c, d)$ is given by $ad - bc$. For the given matrix $(3, -1; 2, k)$, the determinant is $(3)(k) - (-1)(2) = 3k + 2$. For the matrix to be singular, the determinant must be zero: $3k + 2 = 0$. Solving for k: $3k = -2$, which gives $k = -2/3$.

43 [Challenging] • Target: More exponential equations • Cross-Lesson: 5.02 The Dot Product

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

a. 6

b. **8** ✓

c. 9

d. 5

The exponential function is given by $f(x) = 2^x$. To find the value of the function when $x = 3$, we substitute 3 for x into the function's formula: $f(3) = 2^3$ $2^3 = 2 \times 2 \times 2 = 8$ Therefore, the value of the function $f(x) = 2^x$ when $x = 3$ is 8.

44 [Challenging] • Target: Systems and applications • Cross-Lesson: 5.03 Parametric Equations

What is the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$?

a. 4

b. **8** ✓

c. 6

d. 16

To find the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$, we substitute $x = 3$ into the function: $f(3) = 2^{3+1}$ $f(3) = 2^4$ $f(3) = 16$ Therefore, when $x = 3$, $y = 16$.

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45 [Challenging] • Target: Combined logarithmic and exponential practice • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Solve: $e^{2x} = 5$. Express in terms of $\ln$.

a. $x = \ln k**2 + 1 - k$

b. $x = \ln \frac{k**2+1}{k}$

c. $x = \frac{\ln k**2+1}{k}$ ✓

d. $x = \frac{k}{\ln k**2+1}$

Applying Exponential & Logarithmic Functions principles, $2x = \ln 5 \Rightarrow x = \ln 5/2$. Therefore, the verified correct option is $x = \frac{\ln k**2+1}{k}$.

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